than teaching React from scratch, we focus on reviewing crucial concepts that are likely to arise in interview scenarios, either as direct questions or as topics of discussion during coding exercises.

React is built on core concepts that form the foundation of its component-based architecture and efficient rendering system. These include component-based architecture, which promotes reusability and maintainability by enabling modular UI components, and JSX (JavaScript XML), which allows developers to write HTML-like code within JavaScript. React's virtual DOM enhances performance by minimizing direct DOM manipulations, while its declarative UI approach lets developers describe the UI based on the current state rather than imperatively updating the DOM. One-way data binding ensures data flows from parent to child components, and state management can be handled locally, lifted to common ancestors, or managed globally using tools like Redux or Context API. React also offers a rich suite of hooks (see Appendix A.4) that empower functional components to manage state, side effects, and context without using classes. Advanced features like concurrent rendering, suspense for data fetching, and features introduced in recent versions (such as server components and concurrent mode) further enhance React's ability to build fast, responsive, and scalable applications. Together, these features and concepts make React a powerful and flexible library for building modern user interfaces.

While React is currently the most popular framework, developers familiar with other frameworks can use their existing knowledge to learn React quickly. It is always beneficial to understand the key feature differences between React, Angular, and Vue, shown in table 1.1.

Table 1.1 Feature comparison of React, Angular, and Vue

Feature	React	Angular	Vue
Architecture	React is a library focused on UI components.	Angular is a full-featured framework with comprehensive tooling.	Vue is a progressive framework that offers flexible architecture.